LLM and Graphs for Financial Texts

IEORE4737 - AI Applications in Finance

European Investment Bank - EIB 1

Summer 2026

Columbia University

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Contents

Abstract

This report documents the full lineage of the EIB Knowledge Graph project across five consecutive Columbia University teams (Spring 2025 through Summer 2026), building on a 2024 precursor. It covers the data, the extraction and evaluation pipeline, the graph neural network used for link prediction, the recent multi-model LLM benchmark, and the interactive web application built this summer.

1 Executive summary

- **Business question.** How to turn the daily firehose of financial news into a structured, searchable view of what companies, sectors, and instruments are affecting each other over time.
- **Approach.** Large language models (LLMs) read each news article and extract *triplets* — short factual statements of the form "*Nvidia — competes with — AMD*". Triplets accumulate into a knowledge graph. A graph neural network trained on the graph then predicts which entity relationships are likely to appear next.
- **Five semesters of work** (building on a 2024 Llama-based precursor by Yang, Lin & Couzinet):
  - Spring 2025 extracted the first triplets from Yahoo Finance headlines (Llama3-8B), built entity/relation canonicalisation, interactive temporal visualisation, and the first GNN comparison (GAT vs. GraphSAGE).
  - Summer 2025 upgraded the prompt to the 6-tuple format with 14 entity types, invented the handcrafted + Judge-LLM quality metrics, and closed the loop with a third-LLM triplet regeneration pipeline. Expanded the dataset from headlines to full articles (9× longer inputs).
  - Fall 2025 moved to the FNSPID semiconductor corpus and a locally hosted Qwen 2.5 32B, unified the pipeline end to end, and reached MRR 0.8018 with strict semantic filtering — showing cleaner graphs beat larger ones.
  - Spring 2026 hardened the evaluation: fixed a data-leakage bug, showed full article text beats summaries, added causal-feature augmentation that stabilised the model through Covid-19.
  - Summer 2026 (current) added: multi-LLM benchmark (Claude, GPT, Qwen, Gemini) with automated quality scoring; a public, deployed web application with predictions, live corpus, and factor model; and the VM-to-UI publishing pipeline.
- **Current state.**
  - Live web application at https://eib-knowledge-graph.vercel.app with four browsable corpora: the GPT-5.5 refined 2019–2020 corpus (the default source, carrying four directly comparable model encoders — GAT with and without edge features, GraphSAGE, GAT + NOTEARS causal — MRR 0.863–0.942 over 23 backtest months), the finalised full 15-year FNSPID corpus (54,563 articles, 2009–2024, MRR 0.631 over 168 backtest months at ≈466 candidates per query), and two daily-retraining live corpora (STOXX Europe 600 and the full S&P 100) with the news factor model.
  - Controlled data-quality result: holding the model fixed, GPT-5.5 judging/refinement/gating lifts MRR from 0.766 to 0.863 on identical articles.

- Multi-model LLM benchmark completed for four models on identical inputs, scored on four quality dimensions by an independent Judge LLM (Qwen 0.843 vs. GPT-5.5 0.846).
- Fully autonomous production: the entire chain runs on local Qwen at $0 API cost, the live corpora refresh every morning with no human in the loop, and an MCP connector plus a two-tier handover document make the system inheritable.

2 Project timeline

2.1 Spring 2025 — First triplets, canonicalisation, first GNNs

Kaidi Chen; Haining Han; Minhao Pang.

- Collected financial news headlines from the Yahoo Finance RSS feed; extracted subject–predicate–object triplets with Llama3-8B using a 5-tuple prompt — `(entity1, type1, relation, entity2, type2)` — with 11 entity types, five few-shot examples, directional verbs required, dates/numbers excluded.
- Entity-level cleaning: regex filters for temporal patterns (years, quarters, dates), low-frequency entity pruning (<5 occurrences), rule-based validity checks (bare numbers, symbols, fiscal abbreviations).
- Entity & relation canonicalisation: TF-IDF vectorisation (character + word n-grams, 1–4) → cosine similarity → MiniLM-L6-v2 semantic embeddings with sentiment-aligned merging for relations → union-find clustering with canonical label assignment. The direct ancestor of every later dedupe stage.
- Dynamic visualisation: NetworkX + Pyvis interactive graphs, sliceable weekly / monthly / yearly (Figure 1).
- First GNN comparison — GAT vs. GraphSAGE, static vs. rolling windows (Table 1): GraphSAGE static led with MRR 0.550; rolling windows showed adaptability to temporal market shifts.

Model	MRR	Hits@1	Hits@3	Hits@10
GAT Static	0.351	0.176	0.363	0.952
GAT Monthly Rolling	0.444	0.242	0.533	0.966
GAT Quarterly Rolling	0.430	0.220	0.524	0.968
GraphSAGE Static	**0.550**	**0.351**	**0.669**	**0.997**
GraphSAGE Monthly Rolling	0.482	0.267	0.605	0.974
GraphSAGE Quarterly Rolling	0.497	0.284	0.624	0.977

Table 1: Spring 2025 GNN comparison (from the team's poster).

2.2 Summer 2025 — Metrics, Judge LLMs, regeneration

Gunn Lee; Jarvis Gao; Xinnan Lu; Andrea Cuadros.

- Prompt engineering overhaul: 5-tuple → **6-tuple** output adding a relationship-category type (GFMK / SSI / GMM); entity taxonomy 11 → 14 types (adding DERIVATIVE, EQUITIES, MACRO_INDICATOR) with refined definitions; few-shot examples 5 → 8 with errors amended.

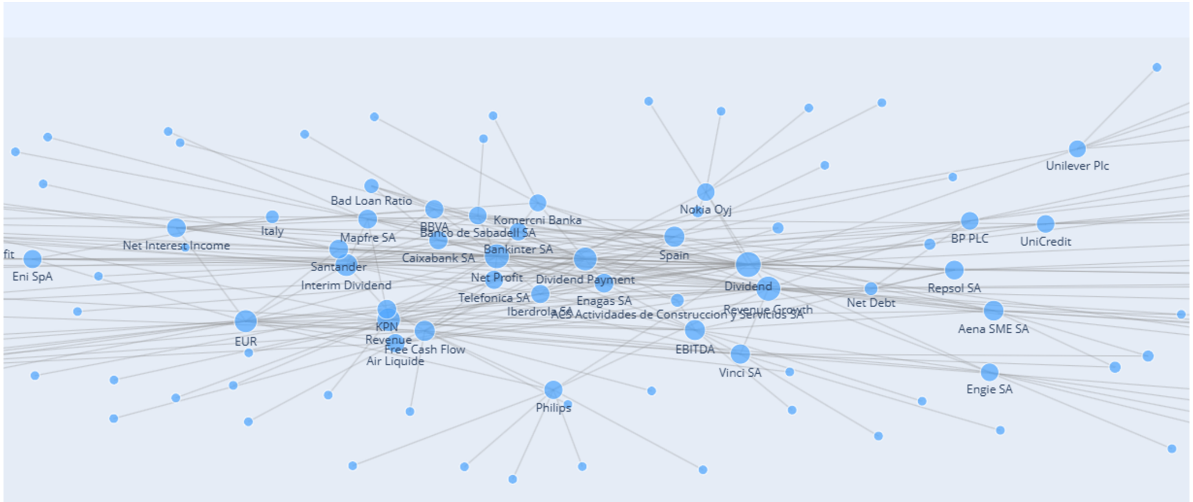

Figure 1: Spring 2025: interactive Pyvis knowledge graph over European financial news — companies, indicators and countries linked by extracted relations.

- **Handcrafted metrics** (unsupervised, lightweight): Coverage (MiniLM-L6-v2 embeddings, weighted cosine similarity), Semantic Uniqueness (similarity grouping; score = groups/triplets), Semantic Validity (ontology domain/range checks), Entropy Ratio (triplet vs. article information density; want < 1).

- **Judge LLMs**: one independent judge per metric dimension, scoring 0–1 with a written explanation for interpretability.

- **Triplet regeneration pipeline** — the origin of the project's three-LLM design: if any LLM metric falls below threshold, a third LLM (Gemini 2.5 Flash) regenerates the triplets from the original article plus the judge and handcrafted evaluations; handcrafted-below/LLM-above disagreements are flagged for human review.

- Dataset expansion: from ≈19k headline-only samples (≈82 words each) to ≈11k full-article samples (≈743 words) — over 9× more context per sample.

- Model comparison on identical inputs (Table 2): Gemini 2.5 Flash beat Amazon Nova Pro on coverage (LLM coverage 0.893 vs. 0.735), comparable on uniqueness and validity.

Metric	Gemini 2.5 Flash	Amazon Nova Pro
Handcrafted Coverage	0.823	0.799
LLM Coverage	0.893	0.735
Handcrafted Uniqueness	0.584	0.768
LLM Uniqueness	0.941	0.974
Handcrafted Semantic	0.890	0.775
LLM Semantic	0.985	0.954
Entropy Ratio	0.735	0.679

Table 2: Summer 2025 extraction-model evaluation, averaged over 10 samples per model (from the team's poster).

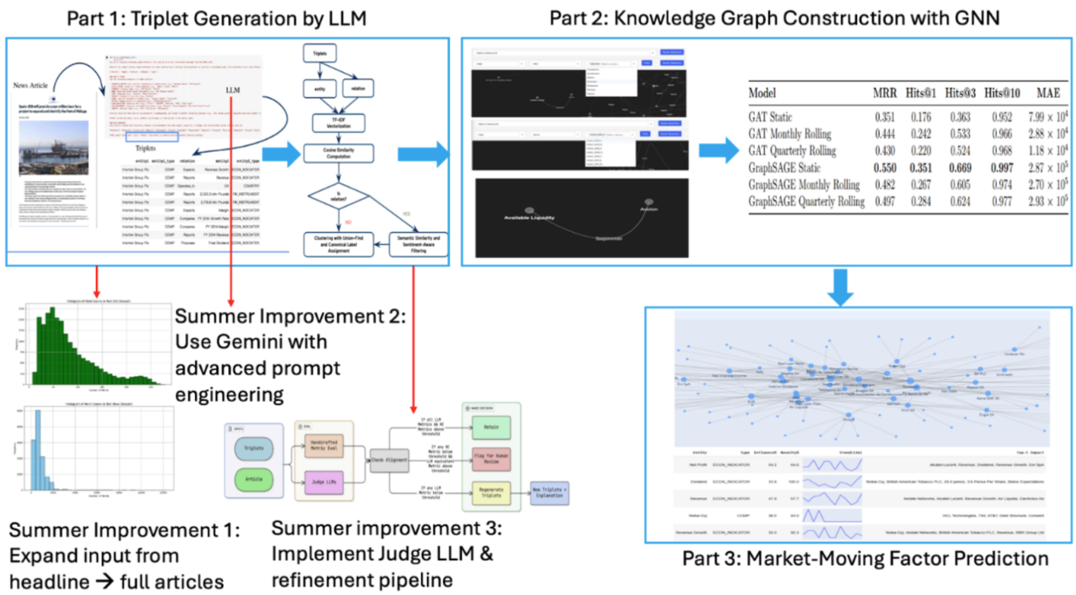

Figure 2: Overview of the pre-Fall-2025 pipeline (from the Fall 2025 poster): triplet generation, knowledge-graph construction with GNNs, and market-moving factor prediction, with the three Summer 2025 improvements annotated.

2.3 Fall 2025 — FNSPID, local Qwen, unified pipeline

Lee, G.; Gao, J.; Lu, X.; Cuadros, A.

- Ingested the FNSPID financial news dataset (Zihan1004 on Hugging Face).
- Restricted focus to semiconductor-related news (2020–2024).
- Extracted triplets using a locally-hosted Qwen 2.5 LLM via Ollama.
- Reverse-engineered the four Judge-LLM quality criteria into the initial extraction prompt itself — shortcutting a slow generate-then-critique loop.
- Defined the four quality dimensions: Coverage, Uniqueness, Semantic Validity, Consistency.
- Built the first Graph Attention Network (GAT) for link prediction; trialled five loss functions (BPR, InfoNCE, Cross-Entropy, FindKG, Hybrid).
- Defined three graph-construction strategies keyed to Judge output: *Original*, *Fallback*, *Strict*.
- Unified the pipeline end to end (Figure 3): article-level handcrafted + Judge evaluation with regenerate-or-flag routing, then corpus-level frequency filtering, TF-IDF similarity clustering, union-find canonical mapping, relation-label consolidation and global dedupe.
- Switched to a locally hosted Qwen 2.5 32B via Ollama with a structured "Mental Sandbox" prompt — rapid iteration, stable throughput, zero API cost.
- Key finding: **cleaner graphs beat larger ones**. The Strict graph, ≈58% smaller than raw, reached the project's best MRR of 0.8018 (BPR loss, no edge features), improving Hits@1

by ≈10%; ranking losses beat classification losses; topology carried more signal than relation semantics.

- Delivered interactive topology-aware visualisation of the monthly graphs (Figure 4) plus global topology metrics tracked over time.

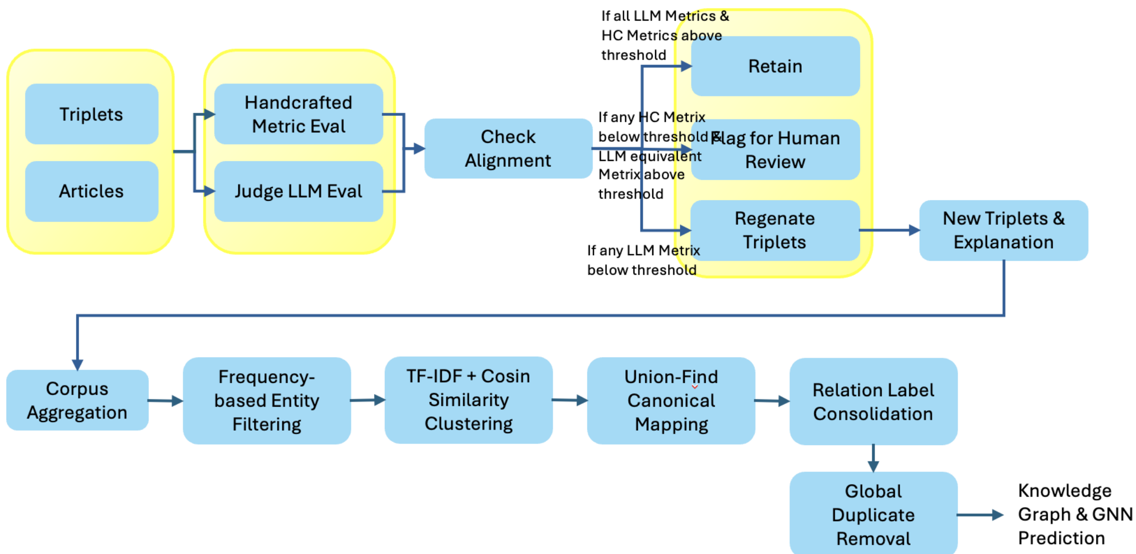

Figure 3: Fall 2025 unified end-to-end pipeline: evaluation-gated retention / human review / regeneration at the article level, then corpus-level canonicalisation into the graph used for GNN prediction.

2.4 Spring 2026 — Robustness and causality

Xixi Chen; Xinyi Chen; Yi Yang.

- Systematically compared triplet extraction from *full article text* versus *article summary*. Full text won by ≈ 0.08 MRR — coverage dominates redundancy.

- Identified and fixed a temporal data-leakage bug in the rolling-window evaluation: the original code built its global entity index from the full dataset before splitting into train/test, so the model implicitly saw future entities through its embedding table.

- Ran a window-size ablation: 3-month training window beat 6-month and 12-month across every configuration — recency dominates historical depth in a non-stationary market.

- Added *causal feature augmentation*: encoded each entity's causal role (in-degree, out-degree, PageRank on a NOTEARS-learned DAG) as a 5-dimensional add-on to the 33-dimensional baseline features.

- Case-studied two market events: Covid-19 (gradual, 2019–2020) and the Russia–Ukraine war (sudden, 2021–2022).
  - Covid-19: causal features cut the shock-period MRR drop by 96%.

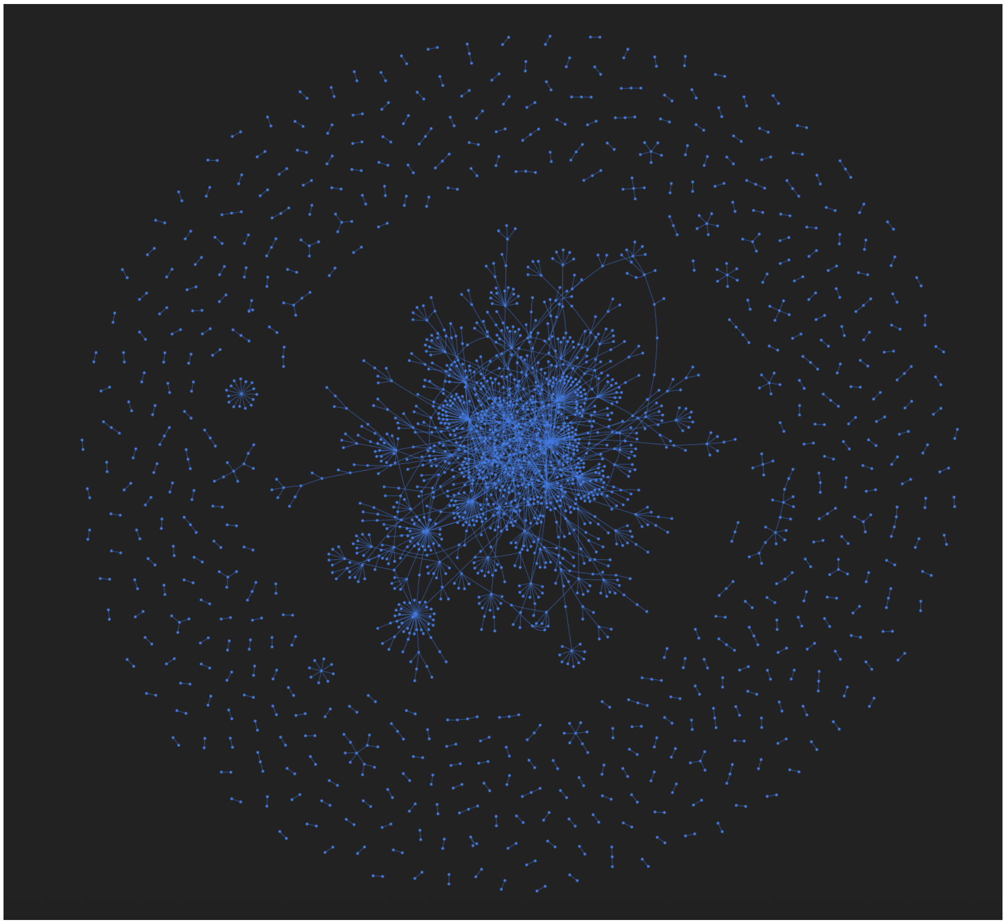

Figure 4: Fall 2025: full knowledge graph over the FNSPID semiconductor corpus — a dense connected core surrounded by small satellite stories.

  - Ukraine war: absolute performance higher; within-shock variance dramatically lower.

- Consolidated the pipeline into a documented package with a Rolling Window subfolder for the corrected training code.

2.5 Summer 2026 — Comparison, UI, deployment

Alexandra Paiz; Pierre Pujol (GitHub SigmaWave); Ruiwen Wang.

- Ran a controlled multi-model LLM benchmark on identical inputs: Claude Sonnet 5, GPT-5, Qwen 2.5 14B, Gemini 2.5 Flash. All extractions scored by the same Judge LLM (Qwen 2.5 14B via Ollama) on Coverage, Uniqueness, Semantic Validity, Consistency.
- Retrained the GAT on the 2019–2020 corpus with the Spring 2026 rolling-window fix; produced 230 checkpoints (10 model configurations × 23 monthly validation periods).
- Built an interactive, publicly deployed web application (Next.js 16, Cytoscape.js, Tailwind CSS, shadcn/ui) at `https://eib-knowledge-graph.vercel.app`.
- Added a FinDKG-style "Top KG Entities" predictions leaderboard driven by real GAT embed-

dings, per month, exposing rank percentile, novelty z-score, 3-month trend sparkline, and top predicted-impacted financial entities.

- Deployed a self-updating news scraper framework (Google News RSS + regulator RSS feeds; GitHub Actions cron; per-corpus watchlists).
- Deployed static builds to Vercel with automatic redeploy on every push to `main`.
- Stood up a fully automated **live corpus** — STOXX Europe 600, 30 curated tickers, refreshed daily by GitHub Actions with zero manual steps (Section 11).
- Built a **news factor model** (five factors + PCA + KMeans archetype clustering) over the live corpus, exposed as a dedicated UI tab (Section 14).
- Brought the extraction up to the three-LLM standard of the team pipeline: judge-and-refine pass plus cross-article entity canonicalisation (Section 13).
- Built the **VM-to-UI publishing pipeline** and used it to ship the full 2-year Qwen extraction (5,358 articles / 30,861 triplets) to the public UI (Section 15).
- Launched the **full 15-year FNSPID corpus run** (54,563 articles, 2009–2024) through the complete three-LLM chain on the team GPU VM at zero API cost, in parallel checkpointed lanes, finalised end to end — canonicalisation, strict graph, CE GAT, live predictions (Section 15).
- Migrated the live corpora from Gemini to **local Qwen on the VM's daily cron** (zero marginal cost) and added a second live corpus: the **full S&P 100** (Section 12).
- Added daily **GAT retraining and predictions for the live corpora** (STOXX MRR ≈0.89 over its first backtest months — provisional, the figure moves with each day's retrain).
- Shipped an **analyst MCP connector** — Claude can be pointed at the deployed corpus and interrogate the news behind any node, factor or prediction (Section 16).
- Reopened the two lineage model questions as directly comparable variants on the GPT-5.5 corpus: **GAT vs. GraphSAGE** (Spring 2025's architecture comparison) and **baseline vs. NOTEARS causal features** (Spring 2026's method, ported verbatim), delivered as a Model encoder toggle with a metric comparison table in the UI. Results: GraphSAGE 0.942 vs. GAT 0.863–0.874; causal features metric-neutral (0.864).
- Team parallel tracks: Pierre judged the full 2-year corpus and ran the 10-configuration GAT sweep on the VM (CE No-Edge-Feats winner, MRR 0.7558); Ruiwen built a GPT-5.5 judge-and-refinement variant (V7) over the 2019–2020 slice via the OpenAI Responses API (≈$150 / 8 h), giving the team a paid-frontier vs. free-local comparison on identical data.

3 Data

3.1 Source corpus

- **FNSPID** (Financial News and Stock Price Integration Dataset), Hugging Face `Zihan1004/FNSPID`.
- 15.7M financial news articles, 1999–2023, mapped to NASDAQ tickers.
- Full articles include title, body, ticker, timestamp, URL.

3.2 Working subset for this project

- **Semiconductor slice** used through Spring 2026: ≈ 2,300 articles across 2019–2020, chosen for its coverage of a full pre-shock / shock / post-shock cycle (Covid-19).
- Second slice used for the Spring 2026 causal study: ≈ 2021–2022, spanning the Russia–Ukraine invasion.
- **Full working corpus (late Summer 2026):** the complete semiconductor extract of FNSPID — 54,563 articles, Dec 2009 – Jan 2024, 11 tickers (NVDA, AMD, INTC, QCOM, MU, TSM, TXN, AMAT, ADI, MRVL, ON) — now processed end to end on the team VM (Section 15). Note the upstream FNSPID release is a static research artefact: its news ends 9 Jan 2024 and the Hugging Face repository has had no commits since the paper's release, which is why the live corpora (Sections 11, 12) exist as the project's forward-moving data.

3.3 Extracted knowledge graph

- File: `summary_triplets_19_20.csv` (Spring 2026 deliverable).
- ≈ 11,800 triplets after quality filtering.
- ≈ 8,000 unique entities across the 24 monthly snapshots.
- Entity types normalised to a closed vocabulary of 18 categories (company, stock ticker, sector, financial instrument, macro indicator, etc.).

4 Pipeline overview

Four sequential stages, unchanged in intent since Fall 2025, refined in every semester:

1. **Triplet extraction.** LLM reads each news article, returns a list of $(\text{entity}_1, \text{type}_1, \text{relation}, \text{relation category},$ tuples.
2. **Quality evaluation.** An independent Judge LLM scores each extraction on four dimensions and, if flagged, requests a revision.
3. **Graph construction.** Triplets become edges of a directed graph; entities become nodes. One graph per calendar month.
4. **Link prediction.** A Graph Attention Network is trained on trailing months and evaluated on the next month — the "predictable" relationships become the model's output.

5 Stage 1 — Triplet extraction

5.1 The triplet as unit of knowledge

- A triplet is one factual claim: "*Taiwan Semiconductor — manufactures for — AMD*".
- Six-slot format: $(e_1, \text{type}_1, r, c, e_2, \text{type}_2)$ — two entities, their types, the relationship verb, and the relationship's high-level category.
- Relationship categories collapse the hundreds of verbs into three families:

- *SSI* — Sector-Specific Insights (product launches, R&D, deals).
- *GMM* — Global Market Movements (macro drivers, cross-market flows).
- *GFMK* — General Financial Market Knowledge (ownership, ticker mapping, structural facts).

- Rationale: keep the vocabulary permissive at the verb level while giving downstream models a coarse handle on relationship meaning.

5.2 Prompt engineering

- Prompt template lives in `components/tgp.txt` (triplet generation prompt).
- Instructs the model to: identify entities, classify each entity type against the ontology, propose the relationship verb, tag the category, avoid duplicates.
- The Fall 2025 team pre-loaded the four Judge criteria into this prompt — so the extractor already tries to satisfy Coverage / Uniqueness / Validity / Consistency in a single pass.

5.3 Models used across semesters

- **Fall 2025:** Qwen 2.5 (14B and 32B variants) locally via Ollama.
- **Spring 2026:** Same Qwen 2.5, focus was downstream not on the extractor.
- **Summer 2026:** Multi-model comparison — Qwen 2.5 14B (baseline), Claude Sonnet 5, GPT-5, Gemini 2.5 Flash.

5.4 Multi-LLM benchmark (Summer 2026)

- **Design.** Same 127-article slice of the corpus fed to each model; identical prompt; identical Judge LLM (Qwen 2.5 14B) scoring the outputs. Fully apples-to-apples.
- **Cost + speed.**
  - Gemini 2.5 Flash: 129 articles, 1,213 triplets, ≈$0.05, 18.7s / article average. 100% well-formed output.
  - Claude / GPT / Qwen: costs and speeds recorded in Pierre's benchmark deck.
- **Robustness.** Signal.SIGALRM per-article timeout (60s) added to prevent silent hangs; incremental CSV checkpointing every 10 articles so an aborted run resumes cheaply.
- **Results** (see Section 6.4) — Gemini 2.5 Flash's Judge scores: Coverage 0.74, Uniqueness 0.83, Semantic Validity 0.84, Consistency 0.92 on 127 articles.

6 Stage 2 — Judge LLM quality evaluation

6.1 Motivation

- Raw LLM output is noisy: hallucinated entities, mis-typed entities, redundant triplets, contradictions.

- Feeding noise into the graph pollutes downstream training.
- Solution: a second LLM (the "Judge") re-reads each extraction and scores it.
- Judge is independent from the extractor to avoid confirmation bias.

6.2 The four dimensions

Coverage — fraction of salient content in the source article captured by the triplet set.

$$\text{Coverage}(a) = \frac{|\{k \in \mathcal{K}_a \mid k \text{ referenced by some } t_i \in \mathcal{T}_a\}|}{|\mathcal{K}_a|}$$

where $\mathcal{K}_a$ is the set of salient semantic units in article a and $\mathcal{T}_a$ is the extracted triplet set.

Uniqueness — penalises redundant triplets (measured via cosine similarity in embedding space):

$$\text{Uniqueness}(a) = 1 - \frac{2}{n(n-1)} \sum_{i<j} \nVdash(\cos(\phi(t_i), \phi(t_j)) > \tau)$$

Semantic Validity — fraction of triplets that are semantically well-formed against the financial ontology:

$$\text{Validity}(a) = \frac{1}{n} \sum_{i=1}^{n} \nVdash(t_i \text{ is semantically valid})$$

Consistency — absence of pairwise contradictions:

$$\text{Consistency}(a) = 1 - \frac{|\{(t_i, t_j) \mid t_i \perp t_j\}|}{\binom{n}{2}}$$

All four are in $[0, 1]$; higher is better.

6.3 Three filtering strategies

- **Original.** Raw LLM output, no Judge intervention. Maximum recall, most noise.
- **Fallback.** Judge's revision applied where valid; falls back to the original if the revision is malformed. Balanced.
- **Strict.** Only triplets explicitly validated or corrected by the Judge kept. Maximum precision, smallest graph.

6.4 Cross-model results (Summer 2026 benchmark)

Gemini 2.5 Flash on the 127-article shared slice:

Dimension	**Mean**	**Median**	**Std**	**Min**	**Max**
Coverage	0.74	0.70	0.07	0.60	0.90
Uniqueness	0.83	0.80	0.10	0.60	1.00
Semantic Validity	0.84	0.85	0.07	0.60	1.00
Consistency	0.92	0.90	0.08	0.70	1.00

Full comparison table (Claude / GPT / Qwen / Gemini side-by-side) — see Pierre's benchmark deck, EIB Report 4.

7 Stage 3 — Knowledge graph construction

7.1 From triplets to graph

- Nodes: distinct entities across all triplets in a given time window.
- Node attributes: entity type (company, sector, ticker, ...).
- Edges: directed, from e_1 to e_2. Attributes carry the verb and its category (SSI / GMM / GFMK).
- Edge weight: multiplicity — how many articles support this exact relationship in the window.

7.2 Temporal snapshots

- One graph per calendar month — 24 snapshots for the 2019–2020 corpus.
- Each snapshot is a self-contained slice: nodes and edges only from articles dated that month.
- Design rationale: financial relationships are non-stationary — what mattered in January 2020 is not what matters in April.

7.3 Text vs. summary — Spring 2026 finding

- Two graphs built from the same articles, one from the article body and one from an LLM-generated summary.
- *Coverage* favoured full text substantially (0.778 vs. 0.687).
- *Uniqueness* favoured the summary (0.918 vs. 0.874) — summaries are naturally less redundant.
- Downstream GAT link prediction favoured the full-text graph consistently by ≈ 0.08 MRR across all 10 model configurations.
- Conclusion: breadth beats precision for graph learning. Redundancy in the source is absorbed by the model; missing information cannot be.

7.4 Static layout (Summer 2026)

- Node positions precomputed with `networkx.spring_layout` at snapshot build time.
- Layout is deterministic — the same month always renders in the same arrangement.
- Frontend consumes these positions directly (Cytoscape "preset" layout) so the graph appears at its final view instantly, no settling animation, no drift.

8 Stage 4 — Graph Attention Network

8.1 What the model is doing, in plain English

The prediction model is a two-part architecture: a **GNN encoder** (GAT here; GraphSAGE in the variant comparison below) that turns each entity into an embedding, and a **link-prediction head** — the dot product of two embeddings, trained with a ranking loss — that scores candidate pairs. The head has no parameters of its own, so all learning lives in the encoder; every variant in this report shares the identical head, which is what makes their metrics directly comparable.

- Assign every entity a 256-dimensional vector (its "meaning").
- Update each entity's vector by attending to its neighbours in the graph, weighted by how relevant each neighbour looks.
- After training, similar vectors indicate the model thinks two entities relate to each other.
- "Link prediction" = ask the model: for entity A, which entities B get the highest dot-product score against A? These are its predictions.

8.2 Architecture

Component	Configuration
Input linear	33 $\rightarrow$ 256 (baseline) or 38 $\rightarrow$ 256 (causal)
Node-type embed	256-dim, one per entity type
Node-id embed	256-dim, 0.5 dropout during training
LayerNorm	After summing input + type + id
GATConv layer 1	256 $\rightarrow$ 256, 4 heads concatenated, edge-dim 33
GATConv layer 2	1024 $\rightarrow$ 256, 1 head, mean aggregation
Edge features	rel_emb(16) + cat_emb(16) + $\log(1 + w)(1)$ = 33-dim

8.3 Loss functions compared

- **BPR** (Bayesian Personalised Ranking) — pairwise ranking: pull true edges above sampled negatives.

$$\mathcal{L}_{\text{BPR}} = -\frac{1}{N}\sum \ln \sigma(s(u, v^+) - s(u, v^-))$$

- **InfoNCE** (contrastive) — treat link prediction as a multi-class problem across K negatives.

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(s(u, v^+)/\tau)}{\exp(s(u, v^+)/\tau) + \sum_i \exp(s(u, v_i^-)/\tau)}$$

- **CE** (Cross-Entropy) — absolute link-existence probability.

$$\mathcal{L}_{\text{CE}} = -[y \log p + (1 - y) \log(1 - p)]$$

- **FindKG** — the margin-based loss from the FinDKG paper.
- **Hybrid** — CE + InfoNCE weighted combination.

8.4 Rolling-window evaluation

- Train on months $[t - W, t]$, evaluate on month $t + 1$, roll forward, repeat.
- For each true edge in the test month, sample 50 negatives; measure whether the model ranks the true one above the negatives.

- Metrics: MRR (Mean Reciprocal Rank), Hits@1, Hits@3, Hits@10.
- "Seen" vs. "Unseen" MRR reported separately — edges between known-in-train entities vs. novel entities.

8.5 Spring 2026 data-leakage fix

- Original bug: global entity index built from the entire dataset before splitting, so the embedding table implicitly contained future entities.
- Fix: build index only from the initial training window; update incrementally as the window rolls forward.
- Effect: honest evaluation — previously optimistic MRRs came down; comparisons across window sizes and loss functions became apples-to-apples.

8.6 Spring 2026 window-size ablation

Config	Splits	MRR	H@1	H@3	H@10
3M train / 1M test	46	**0.8126**	0.7274	0.8740	0.9718
6M train / 1M test	43	0.7934	0.6981	0.8561	0.9578
6M train / 3M test	40	0.7891	0.6943	0.8512	0.9534
12M train / 1M test	37	0.7666	0.6727	0.8275	0.9483

- Monotonic decline with longer training windows — recency wins.
- Interpretation: financial knowledge graph structure is non-stationary; older windows drag in patterns from stale regimes.
- Practical implication: a 3-month rolling window costs less compute AND performs best.

8.7 Spring 2026 causal augmentation

- Baseline node features (33-dim): 32-dim name hash + 1-dim log-degree.
- Causal add-on (5-dim): in-degree, out-degree, in-edge-weight-sum, out-edge-weight-sum, PageRank — all computed on a DAG learned via NOTEARS.
- NOTEARS: continuous optimisation for learning directed acyclic graphs from observational data (Zheng et al., 2018).
- Applied to the top-200 most-mentioned entities to keep the co-occurrence matrix tractable.
- Hypothesis: causal roles are more stable than correlation patterns across market regimes.

8.8 Causal augmentation results

- **Covid-19 (2019–2020).** Baseline MRR fell 0.0763 from pre-shock to shock. Causal model fell only 0.0031 — a 96% reduction in shock drop.

- **Ukraine war (2021–2022).** Causal model absolute performance higher across all phases (pre-MRR 0.7290 vs. 0.6853). Within-shock standard deviation drops by $\approx 6\times$ (MRR σ 0.0026 vs. 0.0152).
- **Trade-off.** Causal features add *noise* during stable periods (small MRR sacrifice) in exchange for shock resilience.
- **Interpretation.** Causal features help most when the disruption is preceded by detectable shifts in news coverage (Covid); help less for sudden exogenous shocks with no prior signal (war).

8.9 Summer 2026 rerun

- Retrained the full 10-configuration leaderboard on the 2019–2020 corpus using the Spring 2026 rolling-window fix.
- Ran on CPU (Apple MPS backend crashed with a known Metal assertion on this graph size).
- Wall time: $\approx$ 19 minutes for the full sweep (10 configs $\times$ 23 validation months).
- Winner: **CE (No Edge Feats), MRR 0.7746**.
- Full leaderboard:

Rank	Model	MRR	H@1	H@3	H@10
1	CE (No Edge Feats)	**0.7746**	0.6891	0.8235	0.9463
2	BPR (No Edge Feats)	0.7465	0.6596	0.7938	0.9182
3	InfoNCE (Rel+Cat+NormW)	0.7455	0.6662	0.7869	0.8988
4	InfoNCE (No Edge Feats)	0.7392	0.6595	0.7768	0.8969
5	Hybrid (No Edge Feats)	0.7334	0.6441	0.7828	0.9092
6	CE (Rel+Cat+NormW)	0.7257	0.6357	0.7738	0.9058
7	BPR (Rel+Cat+NormW)	0.7194	0.6302	0.7637	0.9006
8	FindKG (No Edge Feats)	0.7167	0.6184	0.7716	0.9135
9	FindKG (Rel+Cat+NormW)	0.7049	0.6054	0.7622	0.9077
10	Hybrid (Rel+Cat+NormW)	0.7036	0.6083	0.7575	0.8907

- Interpretation of MRR 0.77: on average the true target entity is ranked $\approx$ 1.3 out of 51 candidates. Random baseline would be $\approx$ 0.05.
- Edge features (relationship type + category) do not consistently help — graph topology alone carries most of the signal.
- Weights saved per (loss function, edge-feature setting, month) — 230 checkpoint files.

8.10 Late-Summer 2026: model-variant comparison

Two lineage questions are reopened on the modern, quality-controlled corpus, as directly comparable *model variants* trained and evaluated identically so that differences come from the model alone.

- **Architecture: GAT vs. GraphSAGE.** Spring 2025 compared the two on early headline-only graphs and found GraphSAGE (mean-aggregation, no attention) ahead — GraphSAGE Static reached MRR 0.550 vs. GAT Static 0.351, with rolling variants between (their published table). We revisit the question with a GraphSAGE twin of the team's GATLP (`scripts/sage_model.py`): identical type/node embeddings, LayerNorm/ELU/dropout stack, training loop and losses — only the message-passing layer differs (SAGEConv instead of GATConv; SAGE has no edge-feature pathway, so it trains in the no-edge configuration). The swap is installed from our wrapper without modifying team code.

- **Features: baseline vs. causal (NOTEARS).** The Spring 2026 causal method is ported verbatim (`scripts/causal_features.py`, attributed): per training window, a causal DAG is learned with their pure-numpy NOTEARS over a daily co-occurrence matrix of the top-200 entities (top-k fixed globally across windows, as in their run), and each node gains their 5 causal dimensions — in/out-degree, in/out edge-weight sums, and PageRank on the DAG, max-normalised. A `--causal` flag applies the augmentation at training and inference through the same non-invasive patching.

- **Delivery.** Each variant publishes its own predictions file and registers in a per-source variants index; the UI's Predictions tab exposes a Model encoder switcher and a side-by-side table of MRR, Hits@1/3/10 and backtest months (Section 10) — the Spring 2025 comparison-table format, instantiated with our runs.

Results. All four encoders trained and backtested identically on the GPT-5.5 refined corpus (23 validation months, same dot-product link-prediction head and cross-entropy recipe throughout, so metric differences isolate the encoder):

Encoder	MRR	Hits@1	Hits@3	Hits@10
GAT (Rel+Cat+NormW edge feats)	0.863	0.801	0.909	0.977
GAT (no edge feats)	0.874	0.815	0.918	0.979
GraphSAGE (mean aggregation)	**0.942**	**0.903**	**0.980**	**0.998**
GAT + causal (NOTEARS)	0.864	0.802	0.911	0.975

Three honest findings. First, GraphSAGE wins decisively on this corpus — the same direction Spring 2025 reported on their early graphs, now reproduced on a quality-controlled corpus with a shared head, which strengthens the lineage claim that attention is not what carries this task. Second, edge features do not pay for themselves here: the no-edge GAT slightly outperforms the edge-featured one, consistent with the sweep result that CE No-Edge-Feats was the winning training configuration on the full corpus. Third, the NOTEARS causal features are metric-neutral (0.864 vs. 0.863 baseline): the causal-graph positions add no ranking signal beyond what message passing already captures on this corpus — mirroring Spring 2026's own finding that causal features mattered mainly for stability under regime shocks rather than for average accuracy.

A fourth finding comes from repeating the comparison at other scales. On the full 15-year corpus (168 backtest months, $\approx$466 candidates per query) GraphSAGE again beats the GAT baseline decisively — MRR 0.788 vs. 0.631 — so the mean-aggregation advantage holds on both large corpora, refined and raw. Only the young STOXX live corpus *inverts* the ranking (GAT 0.885 vs. GraphSAGE 0.506 over five provisional months; its causal variant scores 0.828), suggesting attention earns its keep on small, sparse, title-only graphs while mean aggregation wins once the graph is large. Encoder ranking is corpus-dependent, which is itself an argument for shipping the encoders as swappable variants rather than crowning one.

9 Predictions layer (Summer 2026)

9.1 Design

- Modelled on the FinDKG paper's "Top KG Entities" leaderboard (Ago et al., 2024).
- One row per top entity, per month, per column of information — see Section 9.2.
- Backed by the actual winning GAT weights (CE No-Edge-Feats), not synthetic.
- Exposed in the UI as a full-height tab alongside the graph view.

9.2 Columns

- **Top KG Entity.** The most influential entity in the month's graph, per the GAT.
- **Entity Type.** Company / sector / ticker / instrument / indicator (colour-coded chip).
- **Rank Percentile.** 0–100. Position of this entity when all entities are sorted by mean outgoing GAT prediction score.
- **Novelty (z-score).** Standardised measure of how spiking this entity's activity is versus its own historical baseline. Positive = spiking; near-zero = business as usual; negative = quieting.

$$\text{novelty}(e) = \frac{r(e) - \mu_r}{\sigma_r}, \quad r(e) = \text{edges}_{\text{recent 3mo}}(e) - \overline{\text{edges}_{\text{prior}}}(e)$$

- **Recent 3-mo Trend.** Sparkline of edge counts in months $[P-2, P-1, P]$.
- **Predicted Most Impacted Financial Entities.** Top 5 targets by GAT score, filtered to financial entity types.

9.3 Pipeline

- For each validation month with a GAT checkpoint:
  1. Rebuild the month's graph.
  2. Load the CE No-Edge-Feats weights for that month.
  3. Run inference: produce 256-dim embedding per node.
  4. All-pairs dot product $\rightarrow$ score matrix.
  5. Rank entities by mean outgoing score; compute percentile, novelty, trend, impacted list.
- All 23 monthly leaderboards baked into a single 446 KB static JSON at build time.
- UI fetches once, caches in-memory, and re-renders per selected month with zero backend calls.

9.4 Reality check on the data

- Nothing is LLM-generated for these leaderboards. Trend and novelty are recomputed directly from the source CSV; ranking and impacted-entity lists come from real PyTorch weight files ($\approx$ 10 MB each).

- Verified: for Texas Instruments in March 2020, novelty +3.05 and trend [13, 12, 2] both reproduce exactly from the source data.
- Failure modes visible in the output: extraction noise (stray numeric entities), duplicate entities (e.g. "TXN" and "Texas Instruments Inc" treated as separate nodes). These are upstream extraction issues, not prediction-layer issues; the live corpora no longer exhibit them (generation-time blocklists plus the daily corpus-level canonicalisation), while the historical sources retain them as an honest record of the raw extraction.

10 User interface (Summer 2026)

10.1 Design goals

- Non-technical analysts should be able to browse the knowledge graph without training.
- Every claim in the UI must be traceable to a source (article or model output), not a black-box statement.
- Deployable as a static site so it runs from Vercel's CDN with no backend infrastructure to maintain.
- Aesthetically restrained — optimised for reading in a briefing setting.

10.2 Architecture

- **Frontend:** Next.js 16 (App Router) + TypeScript, Cytoscape.js for graph rendering, Tailwind CSS for styling, shadcn/ui for primitives, next-themes for light/dark.
- **Data layer:** static JSON files under `/data/` (index, per-month snapshots, per-month article corpora, GAT predictions).
- **Optional backend:** FastAPI + Python for local pipeline re-runs and chat; not required in production.
- **Hosting:** Vercel personal account, auto-deploy on push to `main`. Domain: `eib-knowledge-graph.vercel.app`.

10.3 Features

- **Interactive knowledge graph.** Pan, zoom, hover for edge tooltips, click a node to focus its neighbourhood (the rest recedes to reduced opacity but stays readable; the viewport pans to the node without changing zoom).
- **Filter rail.** Toggle entity types on/off; toggle relationship categories; minimum-connections slider (proportional to the current view's max degree).
- **Time slider.** Dual-thumb date range. Preset chips: 1M / 3M / 6M / 12M / All. Defaults to the trailing quarter. Timeline slider drives BOTH the graph tab and the predictions tab.
- **Safari-style tabs.** Knowledge graph / Predictions. Only one visible at a time; state preserved on switch (with a Cytoscape lifecycle fix for the fit-viewport race condition).
- **Predictions view.** FinDKG-style table as described in Section 9.

- **Entity search ($\mathcal{C}$K).** Command-palette-style entity picker across the current snapshot.
- **Article search.** Keyword search across the source news corpus (per-month JSON bundles fetched in parallel, scored client-side).
- **Chat panel.** Optional LLM chat over the corpus (backend-only; user supplies their own API key).
- **Physics settings.** Optional d3-force live simulation for the graph, behind a gear icon. Off by default — static precomputed positions are the default view.
- **Theme.** Light mode by default; dark mode toggle in the header.

10.4 Recent UI iterations (Summer 2026, chronological)

- Replaced polarity edge colouring with 8-family causal-type colouring (Okabe–Ito colourblind-safe palette).
- Fixed graph layout to precomputed `spring_layout` positions — no more settle-from-chaos on page load.
- Added dual-thumb date range selector with client-side snapshot merging.
- Added Safari-style tab strip; predictions moved from a bottom drawer into a full tab.
- Removed purple / neon accents; migrated to neutral palette.
- Fixed Cytoscape crash on tab switch: cancelled pending `requestAnimationFrame` on unmount.

10.5 Late-Summer 2026 overhaul

- **Multi-source architecture.** A source catalogue drives a data-source dropdown; each source declares which tabs it supports (FNSPID: graph + predictions; STOXX: graph + factor analysis; VM run: graph). Unsupported tabs are hidden, not greyed. Per-source data lives under `/data/sources/⟨id⟩/`.
- **Timeline redesign.** Months are equal cells of the track (a 1-month selection is a full cell, not a point). The selected range is a visible window block: drag the middle to slide through time (glides with the pointer, snaps to month boundaries), drag the edges to resize, click the track to jump; preset chips 1M/3M/6M/12M/All; default trailing quarter. On the Predictions tab the same track shows the training months (amber) vs. the predicted month (emerald) of the rolling-window GAT.
- **Obsidian-style graph layout.** A real d3-force simulation runs to convergence before render: many-body repulsion, degree-scaled link springs, a collision force guaranteeing zero node overlap, and centring gravity. Deterministic (seeded), so a given view always lays out identically.
- **Readability at scale.** Node and label sizes are zoom-compensated to constant pixel size across corpora; node size normalised to each snapshot's own degree ceiling; permanent labels allocated by a collision-avoiding budget (all names in moderate or multi-cluster views, top-40% in dense single-blob views, every cluster's top entity always named); auto density threshold targets the $\approx$300 most-connected nodes rather than a fixed percentage of the maximum degree.

- **Industry filter (STOXX).** A sector section above the entity-type checkboxes. Sectors are inferred per news cluster from the watchlist company it concerns; default view opens on Pharma with the other five sectors one click away.
- **Factor analysis tab.** As described in Section 14.
- **Dead-control hygiene.** Controls that do not affect a given tab are hidden there (Factor analysis shows neither the filter rail nor the timeline).
- **Report tab.** This report, viewable in-app as an embedded PDF.
- **Factor history scrubber.** The factor tab gained its own daily timeline: a draggable scrubber with per-date ticks, month-boundary ticks and labels, and a thumb that follows the pointer. Scrubbing is stale-while-revalidate — the current chart stays on screen (dimmed) while the next date's bundle loads, out-of-order responses are dropped, and revisited dates swap in instantly from cache.
- **Live badges.** Live corpora show a pulsing badge whose date is the *last actual pipeline run* (read from the factor index, not a clock) — a stalled cron is visible at a glance; the tooltip names the model chain.
- **Phone support.** At phone widths the filter rail becomes a slide-over drawer behind a Filters button, the factor tab stacks (scatter above, cluster cards below), the predictions leaderboard scrolls horizontally with aligned columns, and the header wraps into stable rows.
- **Layout stability.** Switching data sources moves nothing: the source dropdown has a fixed width, the badge lives on the left, the months counter has a reserved slot, and the header never wraps on desktop (verified pixel-identical control positions across all sources).
- **Model encoder switcher.** Sources shipping several trained models get a Model encoder toggle on the Predictions tab plus a comparison table (encoder × MRR / Hits@1 / Hits@3 / Hits@10 / backtest months); the leaderboard, scorecard and explainer follow the selected encoder, a caption under the table states that every row shares the same dot-product link-prediction head and cross-entropy recipe, and the "How to read this table" block documents each encoder in depth (including the NOTEARS causal variant).
- **Source lifecycle.** Superseded sources (the original 2019–2020 FNSPID slice, then the interim Pierre-judged variant) were retired behind their `available` flags as better corpora covered them; the default source is the GPT-5.5 refined corpus, listed first, with a versioned localStorage key so returning visitors land on it once after the reorder.

10.6 Data flow diagram

News articles (FNSPID)	→	Triplet extraction (LLM)
Triplets	→	Judge LLM quality scoring
Scored triplets	→	Monthly graph snapshots
Snapshots	→	GAT rolling-window training
GAT weights	→	Per-month predictions leaderboard
All artefacts (JSON)	→	Static Vercel deployment
		↓
		Browser fetches JSON, renders UI

11 Live corpus — STOXX Europe 600 pipeline (Summer 2026)

11.1 Motivation

- Everything upstream operated on a fixed historical corpus. Question: can the whole chain — fetch, extract, build graph, deploy — run *live*, unattended?
- Chosen universe: STOXX Europe 600 subset. European coverage is directly sponsor-relevant (EIB).
- Scope deliberately small: 30 curated tickers across six sectors (pharma, tech, banks, industrials & autos, energy, consumer & luxury) so a full daily run fits comfortably in a GitHub Actions job.

11.2 Daily pipeline (fully automated)

- Cron on the team GPU VM (`daily_factors.sh`, 07:30 UTC daily), one loop over every live corpus. No human in the loop; verified self-healing over multiple days (auto-commit with pull-rebase retry when other pushes land mid-run). Initially a GitHub Actions job with Gemini 2.5 Flash; migrated to the VM so extraction runs on **local Qwen 2.5 14B at zero marginal cost** (the Actions workflow is retained as a manual Gemini fallback).
- Steps: fetch Google News RSS per ticker → drop stale republished articles (>120 days old — these previously spawned junk one-article months as far back as 2019) → dedupe against a URL ledger → LLM extraction → refinement passes (Section 13) → merge into monthly snapshots → recompute factor bundle → GAT retrain + predictions (below) → commit. The commit triggers the Vercel redeploy.
- **Enriched extraction schema.** Each triplet carries three extra fields beyond subject/relation/object: sentiment $[-1, 1]$, materiality (US-dollar magnitude mentioned with the fact), and an event-type tag. These feed the factor model.
- **Raw triplet persistence.** Every article's exact triplet list is stored per month before aggregation (`triplets/YYYY-MM.json`) — article-level provenance, exact factor recomputation, and a future retraining dataset.
- Index hygiene: months whose snapshot holds fewer than 3 edges are excluded from the timeline (stray misdated articles cannot pollute the axis).
- Known limitation: Google News RSS supplies titles only (≈60 characters of signal per article) and no deep history — the corpus accumulates forward from activation.

11.3 Retroactive snapshot cleanup and live GAT

- Weeks of pre-refinement extraction had left junk entities and name variants baked into the immutable month snapshots. A snapshot cleaner (`scripts/clean_snapshots.py`) rewrites them in place and now runs inside the daily cron: junk-pattern and generic-phrase removal; a watchlist-driven ticker→company pass (AZN.L, LSE:AZN and plain AZN all merge into AstraZeneca); deterministic corporate-suffix and case merges covering dotted forms (BP p.l.c.), ampersand suffixes (Merck & Co.), share classes (Comcast Corp. Cl A), inverted articles (Walt Disney Company (The)) and hyphen variants (Bristol-Myers), merging across entity types when the name difference is itself suffix evidence; then a conservative TF-IDF fuzzy pass with a numeric-literal guard; edges rewritten with weight summation and degrees recomputed.

- Once three month snapshots existed, the daily cron gained a GAT stage: month snapshots are converted to the trainer's CSV shape (one aggregate row per month, weights preserved), a single CE No-Edge-Feats configuration is retrained over all accumulated months, and a fresh predictions leaderboard is published. First STOXX backtests sit around **MRR 0.89** (few validation months; the figure moves with each day's retrain and will regress toward the FNSPID range as the corpus grows). A GraphSAGE encoder variant trained on the same corpus scores 0.506, the reverse of the refined-corpus ranking — see the corpus-dependence note in Section 8.
- The same stage activates automatically for any live corpus when it crosses the 3-month threshold.

12 Live corpus — S&P 100 (Summer 2026)

- US counterpart of the STOXX corpus, added once the live pipeline proved stable: the **full S&P 100 constituent list** (≈101 tickers; GOOG dual-class dropped in favour of GOOGL) across six sectors — Tech, Financials, Healthcare, Consumer, Energy & industrials, Communications & media.
- Same daily chain as STOXX (Google News RSS → enriched Qwen extraction → judge/refine → snapshots → factors → GAT once ≥3 month snapshots exist), driven by the same VM cron loop — adding the corpus was configuration, not new code: a watchlist YAML, a source-catalogue entry, and a sector keyword map.
- The Industry filter became **per-source**: each live corpus declares its own sector order, keyword map and default-checked sector (STOXX opens on Pharma, S&P on Tech). Sector inference runs per news cluster via union-find majority vote; keyword matching normalises punctuation so names like Coca-Cola, Lowe's and T-Mobile resolve, and ambiguous short tickers (V, T, GE, KO . . .) are matched by company name instead of symbol.
- First full-list run captured ≈485 entities / 365 relationships for its opening month.

13 Extraction refinement — three-LLM parity (Summer 2026)

- The team pipeline (Fall 2025 onward) is a **three-LLM design**: (1) a triplet-extraction LLM; (2) a Judge LLM scoring each article's triplets on coverage, uniqueness and semantic validity; (3) a *flagged-triplet refinement* LLM that receives the article, the bad triplets and the judge's verdict, and regenerates better ones.
- The live pipeline initially ran stage 1 only (plus regex filters) — the source of duplicate entities ("ASML" / "ASML Holding" / "ASML Holding N.V.") and nonsense nodes (bare numbers, generic phrases).
- Now added, adapted to the title-only corpus and collapsed for cost:
  - **Judge-and-refine pass** (one Gemini call per article). Drops entities that are meaningless as graph nodes, merges restated facts, normalises names within the article.

- **Cross-article entity canonicalisation** (one call per run). Maps name variants onto one canonical name each, preferring names already in the corpus so new edges merge into existing nodes instead of duplicating them. Tickers are never merged into companies.

- Both passes *fail open*: an API or parse failure keeps the first-pass triplets, so a flaky call degrades quality for a day rather than losing data.

- **Backend abstraction (late Summer 2026).** All LLM stages of the live pipeline route through one module (`scraper/llm.py`) switchable by environment variable between Gemini and local Ollama models. The production cron runs everything on local Qwen 2.5 14B — the full three-LLM chain at zero marginal cost — with Gemini retained as a fallback backend.

- **Paid-frontier comparison (Ruiwen).** In parallel, a GPT-5.5 judge-and-refinement variant ("V7", OpenAI Responses API with structured outputs, 64 resumable shards, billing-safe request cache, hard quality gates with a human-review route) was built over the same 2019–2020 slice: ≈$150 and 8 hours for the two years, versus $0 for the local-Qwen chain. Together with the extraction benchmark (Section 5.4, Qwen 0.843 vs. GPT-5.5 0.846 judge score), this gives the project a measured cost–quality frontier on identical data. The V7 corpus is deployed as the UI's default source, carrying the four-encoder comparison of Section 8.

14 News factor model (Summer 2026)

14.1 Idea

- First-principles question: beyond drawing a graph, what is the *analytical* utility of daily triplets? Answer: treat news coverage itself as a measurable signal — a Fama–French-style factor decomposition of the news, per entity.

- Every company/topic in the corpus is scored on five factors, i.e. five plain questions about its recent coverage:

  - **Attention** — how much is it written about?
  - **Sentiment** — is the tone positive or negative?
  - **Consensus** — do sources agree, or is the story contested?
  - **Novelty** — is it generating its own storylines or only appearing in others'?
  - **Materiality** — how much money do its stories involve (deal sizes, fines, revenue figures)?

14.2 Method

- Factors roll up from the last two months of stored snapshots (a rolling corpus), not just the day's new articles — the daily delta after ledger warm-up is a handful of articles and produced empty bundles until this fix.

- Scores standardised to z-scores; **KMeans** ($k = 5$) clusters entities in the full five-factor space — no sector labels or company information supplied. Clusters read as automatically discovered *news archetypes*.

- Observed clusters are economically sensible: e.g. Novo Nordisk / Shell / Volkswagen form the high-attention–low-consensus group (heavily covered, contested stories); an energy-heavy cluster (BP, Eni) carries the high-materiality–positive-sentiment signature.
- **PCA** projects the five scores to two axes for display only (≈76% of variance in three components); clustering happens in the full space.
- UI: dedicated "Factor analysis" tab — labelled scatter (collision-avoided names, zoom/pan with constant-size labels so overlaps separate on zoom), archetype cluster cards with plain-language signatures, and a collapsible analyst explainer covering what the tab is for, the factors, the clusters and the map, in non-engineer language.

15 VM-to-UI publishing pipeline (Summer 2026)

- The team's production extraction runs on a Google Cloud VM (`eib-central1a`; Ollama + Qwen 2.5 14B; tmux-detached runs over the full FNSPID corpus). Outputs accumulated on the VM with no path to the UI.
- Built `scripts/publish_vm_output.py`: converts the pipeline's output CSVs (article rows with 6-tuple triplets) into the site's data format — monthly snapshots with layout positions, article records, raw triplet provenance — and pushes; the push triggers the Vercel redeploy. Safe re-publish semantics (`--replace`), dry-run mode, tolerant CSV parsing for large files.
- **First production use.** The full 2019–2021 extraction (Qwen 2.5 14B, pushed to the team pipeline repository on 16 Aug) was published as a third UI source: *FNSPID — qwen2.5 VM run (2019–2021).* 5,358 articles, 30,861 triplets, all 25 months browsable.
- Kept separate from the original FNSPID source so the GAT predictions remain consistent with the extraction they were trained on.
- Documented one-time VM setup (write-scoped deploy key + one cron line) in `docs/VM_PUBLISHING.md`; alternatively a repository-side mirror workflow can poll the public pipeline repo for new outputs, requiring nothing from the VM at all.

15.1 Full 15-year corpus run (late Summer 2026)

- Goal: put the *entire* FNSPID working corpus (54,563 articles, Dec 2009 – Jan 2024) through the complete three-LLM chain — extraction, Judge, flagged-triplet refinement — on the VM's single L4 GPU, at zero API cost, using the benchmark-validated Qwen 2.5 14B for every stage.
- **Chunked parallel lanes.** The corpus is split into date chunks (e.g. 2014–2016, 2022-H1) processed by two independent lanes sharing the GPU via `OLLAMA_NUM_PARALLEL=2`, each in its own working copy of the team pipeline (the per-workdir ontology SQLite store forbids sharing). Per chunk: extract (≈9 s/article) → slice the self-checkpointing master CSV → judge + refine (≈25 s/article, the bottleneck) → publish to the UI source. Every stage checkpoints per article, so crashes and pauses lose nothing. A third stream judges the previously extracted 2019–2021 slice concurrently rather than serialising it at the end. Lane drivers are in `scripts/vm/`.
- **Corpus canonicalisation** (`scripts/canonicalize_corpus.py`). Cross-chunk entity unification: character n-gram TF-IDF + cosine-radius nearest neighbours + union–find, merging only

same-type entities that share an anchor word, with a numeric-literal guard (so “10 nm” never merges into “7 nm”); relation consolidation by fuzzy match; frequency filtering. Two grades: UI grade (min-freq 2) for the browsable graph, and `--strict` (min-freq 5, judge-gated rows only) for GAT input.

- **Corpus finalisation.** Once judged, the corpus is finalised in one chain: canonicalise → republish the browsable graph → strict GAT input → single-configuration CE No-Edge-Feats rolling-window training → predictions, with the Predictions tab enabled automatically. The production backtest spans every rolling month of the corpus — an order of magnitude longer than the 23-month 2019–2020 evaluation, and correspondingly harder.

- Training uses `scripts/run_gat_single.py`, which imports the team’s `run_experiment_case` unchanged but trains only the winning configuration (the stock sweep trains all ten — a 10× saving in production).

- Chunk-level progress publishes incrementally, so the public graph grew from 60 to 100+ browsable months over the run’s lifetime rather than appearing all at once.

16 Analyst MCP connector (Summer 2026)

- The deployed app exposes a Model Context Protocol endpoint (`/api/mcp`) so an analyst can connect Claude (claude.ai custom connector, Claude Desktop or Claude Code) directly to the corpus and interrogate, in natural language, the news behind anything the UI shows.

- Seven read-only tools: source catalogue, timelines, monthly graph (entities + relationships with sentiment / event type / materiality), full-text news search, per-entity news lookup, daily factor bundles, and GAT predictions.

- Implementation is a dependency-free stateless JSON-RPC route handler inside the Next.js app — no SDK, no database, no auth (the data is already public). Tools fetch the same static JSON the frontend renders, so answers always match the site; the endpoint redeploys with every push.

- Cost model matches the project’s constraint: the *analyst’s own* Claude does the reasoning, so serving the connector costs the project nothing at inference time.

- Example session: “Why is Novo Nordisk in the attention+ cluster this week? Show me the articles.” Claude calls `get_factors`, then `get_entity_news`, reads the coverage and answers with cited headlines. Connection instructions and example prompts: `docs/MCP_CONNECTOR.md`.

17 Deployment and infrastructure

- **Repository.** `https://github.com/ap4509-columbia/eib-knowledge-graph` (team collaborators: SigmaWave, Ruiwen04).

- **Vercel project.** `eib-knowledge-graph`. Auto-deploy on push to `main`. Aliased domains: `eib-knowledge-graph.vercel.app` + Git-branch preview URLs.

- **Static-only production build.** No backend at runtime. All data ships as JSON files served from the CDN.

- **Self-updating scraper** (live in production for the STOXX and S&P corpora — Sections 11, 12).
  - Sources: Google News RSS (per-corpus watchlists), ECB / ESMA / BoE regulator RSS feeds.
  - Per-corpus watchlists: US semiconductors, EU utilities, EU banks, EU regulators, STOXX Europe 600 (curated subset), full S&P 100 (both in production).
  - Deduplication ledger prevents re-processing.
  - Runs daily as a cron on the team GPU VM (local Qwen, zero API cost) and auto-commits new snapshots; each commit triggers the Vercel redeploy. The original GitHub Actions workflow remains as a manual Gemini-backed fallback.
- **Local backend** (optional). FastAPI + uvicorn on port 8000 for pipeline re-runs and chat during development.

18 Team contributions summary

Team / semester	Members	Key contributions
Fall 2025	Lee, Gao, Lu, Cuadros	• Baseline end-to-end pipeline • Triplet extraction prompt + ontology • Judge LLM design (4 dimensions, 3 strategies) • First GAT for link prediction • Basic HTML visualisation
Spring 2026	Xixi Chen, Xinyi Chen, Yi Yang	• Full-text vs. summary extraction study • Rolling-window data-leakage fix • 3M / 6M / 12M window ablation • Causal feature augmentation via NOTEARS • Covid-19 + Ukraine war event studies

Summer 2026	Alexandra Paiz, Pierre Pujol, Ruiwen Wang	• Multi-model LLM benchmark (Claude / GPT / Qwen / Gemini) • Interactive web application (Next.js + Cytoscape) and Vercel deployment • FinDKG-style predictions leaderboard • GAT retraining with fixed rolling window (MRR 0.7746) • Full 15-year FNSPID corpus through the three-LLM chain on the GPU VM at $0 API cost, with rolling-window GAT predictions served • Corpus canonicalisation + strict GAT-input pipeline • Live corpora (STOXX Europe 600, full S&P 100) with news factor model and daily GAT retrain, on a daily VM cron • Model-encoder comparison on one shared prediction head: GAT (edge / no-edge), GraphSAGE twin, NOTEARS causal port — shipped as a UI toggle with metrics • Analyst MCP connector over the deployed corpus • Pierre: full 2-year corpus judged on the VM; 10-configuration GAT sweep (CE No-Edge-Feats winner, MRR 0.7558); predictions source • Ruiwen: GPT-5.5 judge-and-refinement V7 (Responses API, sharded + billing-safe, hard quality gates; ≈$150 / 8 h for 2019–2020)

19 Deliverables

19.1 Fall 2025

- `summary_triplets_metrics_22.csv` — earlier scored triplet dump.
- Original pipeline modules under `components/`.
- Static per-month HTML visualisations.

19.2 Spring 2026

- Written report: *LLM and Graphs for Financial Texts* (EIB1_Report.pdf, 16 pages).
- Poster.

- `summary_triplets_19_20.csv` — the 2-year triplet corpus this project has used ever since.
- Corrected rolling-window code (`Rolling Window/` folder).
- NOTEARS causal feature pipeline (`causal_files/` folder).

19.3 Summer 2026

- Deployed web application: https://eib-knowledge-graph.vercel.app.
- Source repository: https://github.com/ap4509-columbia/eib-knowledge-graph (public; VM orchestration scripts under `scripts/vm/`).
- Multi-LLM benchmark: 4-model comparison + Judge LLM stats.
- Retrained GAT: 230 checkpoints + leaderboard CSV.
- Full-corpus VM run: judged + refined + canonicalised FNSPID chunks, master extraction CSVs, canonical UI corpus, strict GAT input.
- Predictions JSON: four encoder-variant leaderboard sets on the GPT-5.5 corpus (23 months each, MRR 0.863–0.942) + 168 monthly leaderboards on the finalised 15-year corpus (MRR 0.631) + live STOXX leaderboards (provisional, retrained daily).
- Live corpora: STOXX Europe 600 and full S&P 100, daily VM cron, factor bundles + raw triplet provenance.
- Analyst MCP endpoint (`/api/mcp`) + `docs/MCP_CONNECTOR.md`.
- Ruiwen's GPT-5.5 judge/refinement bundle (V7) for the paid-frontier comparison.
- This report.

20 Future work

20.1 Corpus extension

- *Resolved at $0.* The full 15-year working corpus (2009–2024) was processed through the local-Qwen three-LLM chain on the team VM (Section 15). The earlier ≈$200 API estimate became unnecessary.
- Beyond that: the upstream FNSPID release extends to 1999 and ≈15M articles across all industries — the same chunked-lane machinery scales to any slice of it, GPU-time permitting.

20.2 Cross-industry expansion

- Current focus: semiconductors.
- Watchlists already defined for EU utilities, EU banks, EU regulators, STOXX Europe 600.
- Enables sponsor-relevant coverage (European infrastructure, regulation, banking).

20.3 Sponsor-facing improvements

- Entity aliasing (dedupe "TXN" and "Texas Instruments Inc") — *shipped for the live corpus* via the canonicalisation pass (Section 13); extending it to the historical FNSPID sources is a one-off batch job over the now-persisted raw triplets.
- Currency numeral cleanup (bare numbers like "1840" as CURRENCY entities) — *shipped for the live corpus* via hard filters + the refine pass; pending for historical sources.
- Promote unmodelled entity types (PERSON, COUNTRY, LOCATION, INSTITUTION) to first-class schema types.
- Sentiment scores from Marketaux or equivalent as a second signal alongside triplets.

20.4 Model integration

- Per-edge GAT confidence exposed as an "uncertainty" colour mode on the graph.
- Similar-entity retrieval from the node embeddings in the detail panel.
- Predicted *new* edges (not just ranking known ones) drawn on the graph with a distinct style.

20.5 Additional event studies

- Interest-rate cycles.
- European energy price spike (2022–2023).
- Banking-sector stress episodes.

A Metric definitions — quick reference

- **MRR** (Mean Reciprocal Rank). MRR $= \frac{1}{|Q|}\sum_{q\in Q}\frac{1}{\mathrm{rank}_q}$. Where rank_q is the position of the true target for query q.
- **Hits@k**. Fraction of queries where the true target is in the top k.
- **Coverage**. Fraction of salient article content captured by the extracted triplets.
- **Uniqueness**. $1-$ redundancy rate among the triplets (cosine similarity in embedding space).
- **Semantic Validity**. Fraction of triplets that fit the financial ontology.
- **Consistency**. Absence of pairwise contradictions.
- **Novelty z-score**. Standardised measure of an entity's activity spike vs. its own history.
- **Rank Percentile**. Position of an entity when ranked by mean outgoing GAT prediction score, expressed as a percentile.

B Software stack — full list

- **Extraction:** Python, Ollama (Qwen 2.5 14B / 32B), Google Gemini API (google-genai), Anthropic API, OpenAI API.
- **Modelling:** PyTorch, PyTorch Geometric, NetworkX, SciPy, scikit-learn (PCA, KMeans), SentenceTransformers.
- **Backend:** FastAPI, uvicorn, pandas.
- **Frontend:** Next.js 16, TypeScript, Cytoscape.js, cytoscape-d3-force, Tailwind CSS, shadcn/ui, next-themes, lucide-react.
- **Infrastructure:** GitHub, GitHub Actions (cron), Vercel (deployment, CDN).

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